III: Metrical theory and superoptimal CFs

Approximation coefficients

- Defn: The approx. coeff. of $x \notin \mathbb{Q}$, $p/q \in \mathbb{Q}$ (reduced) is $\textcircled{H}(x, p/q) := q^2|x - p/q|$.
- Recall p_n/q_n are RCF-convergents of x
- Classical results (all constants are optimal):
  - Thm. (Hurwitz, 1891): $\forall x \notin \mathbb{Q}$, $\exists$ infinitely many $p/q \in \mathbb{Q}$ s.t.
    $$\textcircled{H}(x, p/q) < 1/\sqrt{5}.$$
  - Thm. (Legendre, 1798):
    $$\textcircled{H}(x, p/q) < 1/2 \implies p/q = p_n/q_n \text{ for some } n.$$
  - Thm. (Vahlen, 1895): $\forall x \notin \mathbb{Q}$, $n \in \mathbb{N}$,
    $$\min\{\textcircled{H}(x, p_{n+j}/q_{n+j}) \mid j = 0, 1\} < 1/2$$
  - Thm. (Borel, 1903): $\forall x \notin \mathbb{Q}$, $n \in \mathbb{N}$,
    $$\min\{\textcircled{H}(x, p_{n+j}/q_{n+j}) \mid j = 0, 1, 2\} < 1/\sqrt{5}$$

- Doeblin–Lenstra Conj. (Bosma–Jager–Weidijk, 1983): For a.e. x,

$$\lim_n \frac{1}{n} \#\left\{0 \le k < n \mid \textcircled{H}(x, p_k/q_k) < c\right\}$$

$$= \begin{cases} \frac{c}{\log 2}, & 0 \le c \le 1/2 \\ \frac{1 - c + \log(2c)}{\log 2}, & 1/2 \le c \le 1 \end{cases} \qquad \forall c \in [0,1].$$

- Several analogues, refinements of these results have been given for RCF-convergents and mediants

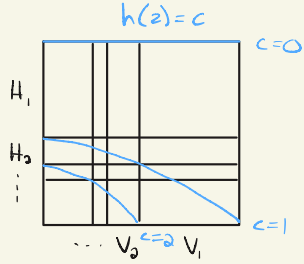
- We give a generalised Doeblin–Lenstra thm. which captures several of these

- Recall: $\mathcal{F}^n(x,y) = \left(A_{[0,n]}^{-1} \cdot x,\ A_1 A_{[0,n]}^{T} A_1^{-1} \cdot y\right)$,

$$A_{[0,n]} = \begin{pmatrix} u_n & r_n \\ s_n & t_n \end{pmatrix} := \begin{pmatrix} \lambda_n p_{n-1} + p_{n-2} & p_{n-1} \\ \lambda_n q_{n-1} + q_{n-2} & q_{n-1} \end{pmatrix}.$$

- Define $h : \Omega \setminus \{(0,0)\} \to [0,\infty)$

$$z = (x,y) \mapsto \frac{1-y}{x+y-xy}.$$



- AC Prop.: $\textcircled{H}(x, u_n/s_n) = h(\mathcal{F}^n(x,1))$.

- Proof: Set $(x_n, y_n) = \mathcal{F}^n(x,1)$. Then

$$x = A_{[0,n]} \cdot x_n = \frac{u_n x_n + t_n}{s_n x_n + r_n} \quad \Longrightarrow$$

$$\textcircled{H}(x, u_n/s_n) = s_n^2 \left| \frac{u_n x_n + t_n}{s_n x_n + r_n} - \frac{u_n}{s_n} \right| = s_n^2 \left| \frac{s_n t_n - u_n r_n}{(s_n x_n + r_n) s_n} \right| \quad (= |\det A_{[0,n]}|)$$

$$= \frac{s_n}{s_n x_n + r_n},$$

Also,

$$y_n = A_1 A_{[0,n]}^{\tau} A_1^{-1} \cdot 1 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} u_n & s_n \\ t_n & r_n \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 0 \end{pmatrix} \cdot 1$$

$$= \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} u_n & s_n \\ t_n & r_n \end{pmatrix} \cdot 0 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \cdot \frac{s_n}{r_n} = \frac{r_n}{s_n + r_n},$$

hence

$$h(x_n, y_n) = \frac{1 - y_n}{x_n + y_n - x_n y_n} = \frac{1 - y_n}{x_n(1 - y_n) + y_n} = \frac{s_n}{s_n x_n + r_n}. \qquad \square$$

- Recall for $R \subset \Omega$, $\frac{u_n^R}{s_n^R} := \frac{u_{N_n^R}}{s_{N_n^R}}$ where

$$\mathcal{F}_R^n = \mathcal{F}^{N_n^R}.$$

- AC Cor.: $\textcircled{H}(x, u_n^2/s_n^2) = h(\hat{\mathcal{T}}_R(x,1))$.

- To understand generic behavior of $\textcircled{H}(x, u_n/s_n)$, suffices to understand $\mathcal{T}^n(x,1)$ for generic x.

- By the ratio ergodic thm., $\forall f, g \in L^1(\bar{u})$, $g \geq 0$, $\int g \, d\bar{u} > 0$,

$$(*) \qquad \lim_n \frac{\sum_{k=0}^{n-1} f(\mathcal{T}^k(z))}{\sum_{k=0}^{n-1} g(\mathcal{T}^k(z))} = \frac{\int f \, d\bar{u}}{\int g \, d\bar{u}} \qquad \bar{u}\text{-a.s.}$$

- Problems:
  (i) $\{z = (x,1)\}$ is a $\bar{u}$-null set
  (ii) $\{z \mid (*) \text{ holds}\}$ depends on f, g

- Brown-Yin (1996) circumvent (i) by imposing 'Hölder'-like conditions on f, g

- Set $\mathcal{A} := \left\{ R \subset \Omega \;\middle|\; \begin{array}{l} R \in \mathcal{B},\ 0 < \bar{u}(R) < \infty, \\ \bar{u}(\partial R) = 0,\ (0,0) \notin \overline{R} \end{array} \right\}$

- Thm. 1 (Dajani-Kraaikamp-S., 2025): For λ-a.e. $x \in (0,1)$,

(**) $$\lim_n \frac{\sum_{k=0}^{n-1} \mathbb{1}_S(\mathcal{F}^k(x,1))}{\sum_{k=0}^{n-1} \mathbb{1}_R(\mathcal{F}^k(x,1))} = \frac{\bar{\mu}(S)}{\bar{\mu}(R)} \qquad \forall S, R \in \mathcal{A}$$

- Idea: Find countable collection $\mathcal{C} \subset \mathcal{B}$ which 'approximate' such S, R 'from above/below'. Then (*) holds $\bar{\mu}$-a.s. for any f, g indicator functions on $\mathcal{C}$. For λ-a.e. x, $\exists y$ s.t. $z = (x,y)$ belongs to this full $\bar{\mu}$-set. Show $|\mathcal{F}^n(x,1) - \mathcal{F}^n(x,y)| \to 0$ uniformly, then squeeze the limit in (**) from above/below with '$\bar{\mu}$-generic (x,y)'.

- Cor.: For a.e. x,

$$\lim_n \frac{1}{n} \sum_{k=0}^{n-1} \mathbb{1}_S(\mathcal{F}_R^k(x,1)) = \bar{\mu}_R(S)$$

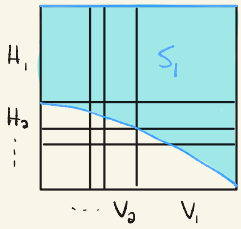
$\forall S, R \in \mathcal{A}$, $S \subset R$.

- Proof: The left-hand side equals

$$\lim_n \frac{\sum_{k=0}^{n-1} \mathbb{1}_S(\mathcal{F}_R^k(x,1))}{\sum_{k=0}^{n-1} \mathbb{1}_R(\mathcal{F}_R^k(x,1))} = \lim_n \frac{\sum_{k=0}^{N_n^R} \mathbb{1}_S(\mathcal{F}^k(x,1))}{\sum_{k=0}^{N_n^R} \mathbb{1}_R(\mathcal{F}^k(x,1))}$$

$$\overset{(\text{Thm. 1})}{=} \frac{\bar{\mu}(S)}{\bar{\mu}(R)} = \bar{\mu}_R(S) \quad \text{a.s.} \qquad \square$$

- For $c \in [0, \infty)$, set $S_c := h^{-1}([0,c))$.



From this cor., we obtain a generalised Doeblin-Lenstra thm.:

- Thm. 2 (D.-K.-S., 2025): For a.e. x,

$$\lim_n \frac{1}{n} \#\left\{0 \le k < n \mid \oplus(x, u_k^R/s_k^R) \le c\right\} = \bar{\mu}_R(S_c \cap R)$$

$\forall R \in \mathcal{A}$ and $c \in [0, \infty)$.

- Proof: By AC Cor., the left-hand side is

$$\lim_n \frac{1}{n} \#\left\{0 \le k < n \mid h(\mathcal{F}_R^k(x,1)) \le c\right\}$$

$$= \lim_n \frac{1}{n} \sum_{k=0}^{\hat{n}} \mathbb{1}_{S_c \cap R}(\mathcal{F}_R^k(x,1)) = \bar{\mu}_R(S_c \cap R) \quad \text{a.s.} \qquad \square$$

- Recall certain R 'announce' particular convergents, e.g.

$$R = V_{a-\lambda} \cap H_{\lambda+1} \implies \frac{u_n^R}{s_n^R} = \frac{\lambda p_j + p_{j-1}}{\lambda q_j + q_{j-1}}, \quad a_{j+1} = a.$$

- Choosing appropriate R, simple calculus (i.e., computing $\bar{\mu}_R(S_c \cap R)$) gives several old and new limiting distributions

Subregion R	Subsequence $(u_n^R/s_n^R)_{n\geq 0}$ of $\left(\frac{u_n}{s_n}\right)_{n\geq 0} = \left(\frac{\lambda_n p_{j_n}+p_{j_n-1}}{\lambda_n q_{j_n}+q_{j_n-1}}\right)_{n\geq 0}$ specified by R	Original statement
$H_{\lambda+1}$, $\lambda > 0$	$\lambda_n = \lambda$	Theorem 1.9 of [20]
V_a, $a > 0$	$\lambda_n = a_{j_n+1} - a$	
$V_a \backslash H_1$, $a > 0$	$\lambda_n = a_{j_n+1} - a > 0$	Theorem 3.2 of [20], when $a = 1$
S_{c_0}, $c_0 \geq 0$	$\Theta(x, u_n/s_n) \leq c_0$	Theorem 2.2 of [20]
$V_{a-\lambda} \cap H_{\lambda+1}$, $0 \leq \lambda < a$	$\lambda_n = \lambda$ and $a_{j_n+1} = a$	Theorem 3.1 of [20], when $\lambda = a - 1 \geq 1$
$\bigcup_{\lambda=0}^{\Lambda} H_{\lambda+1} \cup \bigcup_{a=1}^{A} V_a$, $\Lambda \geq 0$, $A \geq 1$	$\lambda_n \leq \Lambda$ or $a_{j_n+1} - A \leq \lambda_n \leq a_{j_n+1} - 1$	Theorem 4.iii of [29], when $\Lambda = A$; Theorems 3.3, 3.1, 2.20 of [20, 93, 97], resp., when $\Lambda = A = 1$

[20] BOSMA, W. Approximation by mediants. *Math. Comp. 54*, 189 (1990), 421–434.

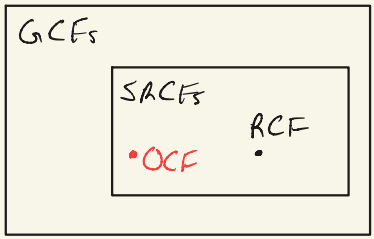
[29] BROWN, G., AND YIN, Q. Metrical theory for Farey continued fractions. *Osaka J. Math. 33*, 4 (1996), 951–970.

[93] ITO, S. Algorithms with mediant convergents and their metrical theory. *Osaka J. Math. 26*, 3 (1989), 557–578.

[97] JAGER, H. Some metrical observations on the approximation of an irrational number by its nearest mediants. *Period. Math. Hungar. 23*, 1 (1991), 5–16.

Superoptimal CFs

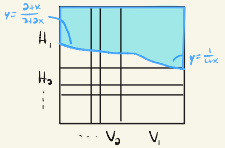
- Def'n.: A semi-RCF (SRCF) is a GCF $[d_0/e_0; e_1/d_1, e_2/d_2, \ldots]$ with
  - (i) $e_n = \pm 1$
  - (ii) $d_n > 0$
  - (iii) $e_{n+1} + d_n \geq 1$, strict i.o.
- Fact: SRCFs always converge
- For given x, have following inclusions of exp'ns:



- Bosma (1987) introduced an algorithm producing optimal CFs (OCF).
- The OCF-exp'n. of x is a SRCF-exp'n. whose convergents P_k/Q_k satisfy
  - (i) $\oplus(x, P_k/Q_k) < 1/2 \quad \forall k$, and
  - (ii) $\lim_k \frac{n(k)}{k} = \frac{\log 2}{\log \phi}$ a.s.,

  where $\phi = \frac{\sqrt{5}+1}{2}$, $q_{n(k)} \leq Q_k < q_{n(k)+1}$.
- Remark: OCFs are S-exp'ns, hence also contracted Farey exp'ns. with

$$R = \left\{(x,y) \in \Omega \mid y > \min\left\{\frac{2+x}{3+2x}, \frac{1}{1+x}\right\}\right\}$$



- The OCF is optimal among SRCFs...

(for a.e. x, the constants in (i), (ii) cannot be improved for any SRCF-expn. of x)

but not necessarily among GCFs

- Defn.: Let $\varepsilon, C > 0$. A GCF-expn. of x is $\underline{(\varepsilon, C)\text{-superoptimal}}$ if both

(i) $\Theta(x, P_k/Q_k) \le \varepsilon \quad \forall k$, and

(ii) $\liminf_k \frac{n(k)}{k} \ge C$.

- Thm. 3 (S. 2026+): Let $R \in \mathcal{A}$ with

a) $R \subset S_\varepsilon$ and b) $\overline{u}(R) \le \frac{\log 2}{C}$.

Then for a.e. x, the CFE of x w/r/t R is (ε, C)-superoptimal.

- Proof:

(i)

$$\Theta\left(x, \frac{P_k}{Q_k}\right) = \Theta\left(x, \frac{U_{k+1}^R}{S_{k+1}^R}\right) \overset{\text{(AC Cor.)}}{=} h\left(T_R^{k+1}(x,1)\right) \overset{\text{(a) and Defn. of } S_\varepsilon}{<} \varepsilon$$

(ii) Set $N = N_{k+1}^R$, and recall

$$Q_k = S_{k+1}^R = S_N = \lambda_N q_{jN} + q_{jN-1},$$

So $q_{j_N-1} \le Q_k < q_{j_N+1}$

$\Rightarrow \quad n(k) = j_N + \delta, \qquad \delta = \delta(k) \in \{0, -1\}.$

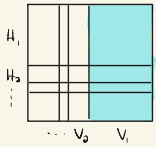
Suffices to show $\lim_k \frac{j_{N_k^R}}{k} \ge C$ a.s.

Recall j_n, λ_n defined by

$$n = a_1 + \cdots + a_{j_n} + \lambda_n, \qquad 0 \le \lambda_n < a_{j_n+1}.$$

One can rewrite

$$j_n = \#\{0 \le k < n \mid \mathcal{T}^k(x,1) \in V_1\},$$



so for a.e. x,

$$\lim_k \frac{j_{N_k^R}}{k} = \lim_k \frac{\sum_{i=0}^{N_k^R-1} \mathbb{1}_{V_1}(\mathcal{T}^i(x,1))}{\sum_{i=0}^{N_k^R-1} \mathbb{1}_R(\mathcal{T}^i(x,1))} \overset{\text{(Thm. 1)}}{=} \frac{\bar{\mu}(V_1)}{\bar{\mu}(R)}$$

$$= \frac{\log 2}{\bar{\mu}(R)} \overset{\text{(b)}}{\ge} C. \qquad \square$$

Legendre CFs

- Fix $\varepsilon_0 \le 1/2$, and suppose $\Theta(x, p/q) < \varepsilon_0$ i.o.

Let $R = S_{\varepsilon_0}$. Since $\{p_n/q_n\} \subset \{U_n/S_n\}$,

Legendre's Thm. and AC Cor. $\Rightarrow$

$$\{U_n^R/S_n^R\} = \{p/q \in \mathbb{Q} \mid \Theta(x, p/q) < \varepsilon_0\}.$$

- Call the CFE w/r/t R the $\underline{\varepsilon_0\text{-Legendre}}$ CF. Since $\bar{u}(R) = \varepsilon_0$, these are a.s. $(\varepsilon_0, \log 2/\varepsilon_0)$ – superoptimal.

(- Remark: Minkowski's diagonal CF is the $1/2$-Legendre CF)

- E.g.: Let $\varepsilon_0 = 1/\sqrt{5}$, and define

$$R_1 := \{z \in R \mid \mathcal{T}_{H_1}(z) \in R\},$$

$$R_j := \{z \in R \setminus \bigcup_{\ell=1}^{j-1} R_\ell \mid \mathcal{T}_{H_1}^j(z) \in R\}, \quad j > 1.$$

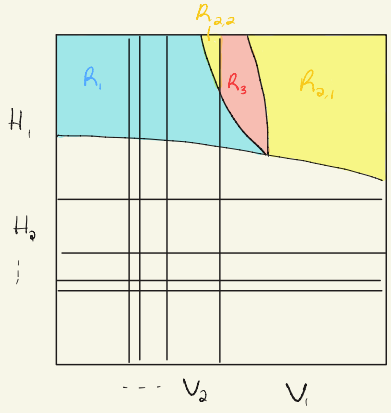
One finds

$$R_1 = \{z \in R \mid \frac{xy}{x+y-xy} < 1/\sqrt{5}\},$$

$$R_2 = \underbrace{\{z \in R \mid \frac{1-x}{x+y-xy} < 1/\sqrt{5}\}}_{R_{2,1} \subset V_1} \sqcup \underbrace{((R \setminus R_1) \cap V_2)}_{R_{2,2} \subset V_2},$$

$$R_3 = R \setminus (R_1 \cup R_2).$$

Since $R_j \neq \emptyset \ \forall j > 3$, this implies Borel's and Hurwitz's thms.



Further, for $z = (x, y) \in R$,

$$A_R(z) = \begin{pmatrix} u_R(z) & t_R(z) \\ s_R(z) & r_R(z) \end{pmatrix} = \begin{cases} \begin{pmatrix} 0 & 1 \\ 1 & a \end{pmatrix}, & z \in R_1 \text{ if } \left\lfloor \frac{1}{x} \right\rfloor = a \\ \begin{pmatrix} 1 & a \\ 1 & a+1 \end{pmatrix}, & z \in R_{2,1} \text{ if } \left\lfloor \frac{x}{1-x} \right\rfloor = a \\ \begin{pmatrix} 1 & a \\ 2 & 2a+1 \end{pmatrix}, & z \in R_{2,2} \text{ if } \left\lfloor \frac{x}{1-2x} \right\rfloor = a \\ \begin{pmatrix} 1 & a+1 \\ 2 & 2a+1 \end{pmatrix}, & z \in R_3 \text{ if } \left\lfloor \frac{1-x}{2x-1} \right\rfloor = a \end{cases}$$

Together with Digits Prop. (Lecture 2), this gives an explizit algorithm to find precisely those $p/q \in \mathbb{Q}$ satisfying Hurwitz's thm. $\forall x \notin \mathbb{Q}$.